

Deloitte Job Simulation

Task 1 Tableau Dashboard

Objective

Analyze one month of machine telemetry data (May 2021) from four Daikibo factories to identify:

1. Which factory experienced the most machine downtime?
 2. Which machine type contributed the most downtime in that factory?
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Daikibo has **4 factories**:

1. **Daikibo Factory Meiyo** → Tokyo, Japan
 2. **Daikibo Factory Seiko** → Osaka, Japan
 3. **Daikibo Berlin** → Berlin, Germany
 4. **Daikibo Shenzhen** → Shenzhen, China
-

What data did they collect?

Each factory has **9 machine types**.

Every machine sends a message every **10 minutes** saying whether it is:

- **Healthy** = working fine 
- **Unhealthy** = problem/breakdown 

Daikibo collected these messages for **one month (May 2021)**.

All the data was stored in one file:

daikibo-telemetry-data.json

Why did they collect this data?

The company wants answers to 2 questions:

Question 1

Which factory had the most machine breakdowns?

Example:

Factory	Breakdown Time
Meiyo	110 mins
Seiko	480 mins
Berlin	20 mins
Shenzhen	420 mins

Answer:

Seiko had the most downtime.

Question 2

Inside that factory, which machine broke the most?

Example:

Machine Type	Downtime
LaserWelder	480
LaserCutter	430
CNC	10

Answer:

LaserWelder broke the most.

In the Deloitte Task

The dataset does not tell us exactly when a machine broke and when it was repaired.

It only tells us:

Status
healthy

Status
unhealthy

Example:

Time	Status
10:00	healthy
10:10	unhealthy
10:20	unhealthy
10:30	healthy

From this we know:

- The machine had a problem between 10:10 and 10:30.
- It was unhealthy for 20 minutes.

So Tableau counts:

10:10 → unhealthy = 10 min

10:20 → unhealthy = 10 min

Total downtime = 20 min

Why "Breakdown Time $\approx$ Downtime"?

Normally:

- Breakdown Time = time the machine is actually broken.
- Downtime = total time the machine cannot be used.

But in this dataset, we only have the status unhealthy.

So whenever the machine is unhealthy, we assume:

Machine unhealthy = Machine down

Therefore:

Breakdown Time $\approx$ Downtime

JSON FILE DATA

The JSON file contains telemetry (machine health) data collected from all 4 Daikibo factories.

Think of it like a huge logbook where every machine sends a message every 10 minutes.

Main fields in the JSON file

Field	Meaning
Device ID	Unique ID of a machine
Device Type	Type of machine (LaserWelder, LaserCutter, etc.)
Factory	Which factory the machine belongs to
Status	Healthy or Unhealthy
Temperature	Machine temperature
Timestamp	Date and time of the message

Example Record

A single row might look like:

```
{
  "deviceId": "M001",
  "deviceType": "LaserWelder",
  "factory": "daikibo-factory-seiko",
  "status": "unhealthy",
  "temperature": 82,
  "timestamp": "2021-05-15 10:10:00"
}
```

Meaning:

- Machine = LaserWelder
- Factory = Seiko
- Time = 10:10 AM
- Status = Unhealthy
- Temperature = 82°C

Since it's unhealthy:

10 minutes downtime

What Tableau Used

We mainly used:

Factory

To answer:

Which factory had the most downtime?

Device Type

To answer:

Which machine broke the most?

Status

To check:

healthy

or

unhealthy

Unhealthy (Calculated Field)

We created:

IF Status = "unhealthy" THEN 10

ELSE 0

END

In simple words

 **If the machine is unhealthy, count 10 minutes of downtime.**

 **If the machine is healthy, count 0 minutes of downtime.**

Example 1

Status

unhealthy

Formula result:

10

Meaning:

Machine was down for 10 minutes

Example 2

Status

healthy

Formula result:

0

Meaning:

No downtime

Why 10?

Because every machine sends a message every:

10 minutes

So:

Status	Downtime
unhealthy	10
healthy	0

Example with 4 records

Time	Status	Formula Result
10:00	healthy	0
10:10	unhealthy	10
10:20	unhealthy	10
10:30	healthy	0

Total:

$$0 + 10 + 10 + 0 = 20$$

So:

Total Downtime = 20 minutes

Why did we create this field?

Tableau cannot directly calculate downtime from "healthy/unhealthy" text.

So we converted:

healthy → 0

unhealthy → 10

Then Tableau could add all the values and tell us:

- Which factory had the most downtime
- Which machine had the most downtime

That's why LaserWelder = 480, meaning Tableau found 48 unhealthy records × 10 minutes = 480 minutes downtime. 

How much data was in the file?

When you loaded it into Tableau, it showed:

160,704 rows

That means:

- Thousands of machine messages
 - From 4 factories
 - Over 1 month (May 2021)
 - Every 10 minutes
-

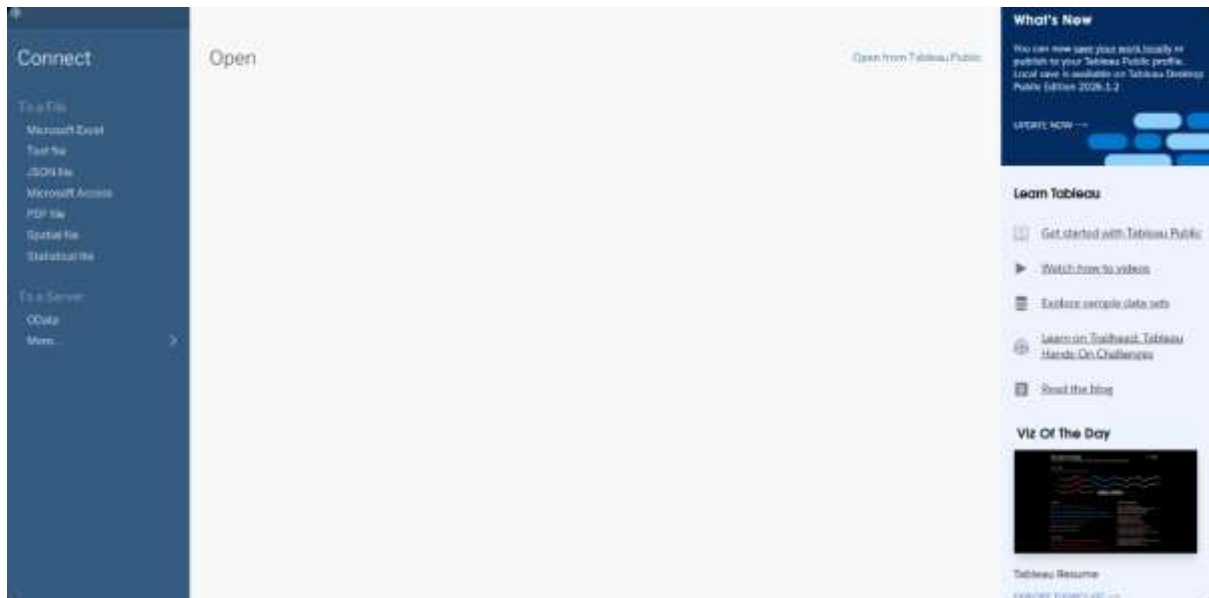
Final Insight from the JSON Data

Using the JSON file, we found:

- Factory with most downtime: Daikibo Factory Seiko
- Machine with most downtime there: LaserWelder
- Total downtime: 480 minutes

So the JSON file is simply a collection of machine health messages that Tableau analyzes to find breakdowns and downtime. 🚀

What did we do in Tableau?



Step 1: Import the JSON file

On the left side, click:

JSON file

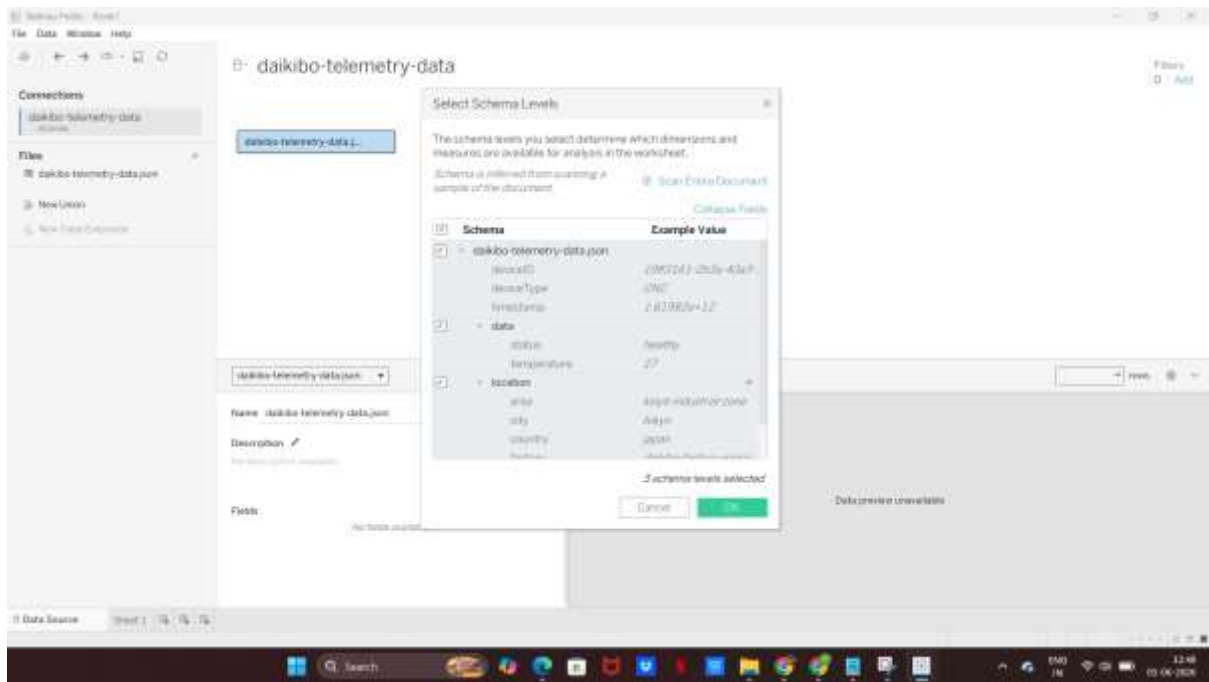
(under To a File)

Step 2

Browse to:

daikibo-telemetry-data.json

Select it and click Open.



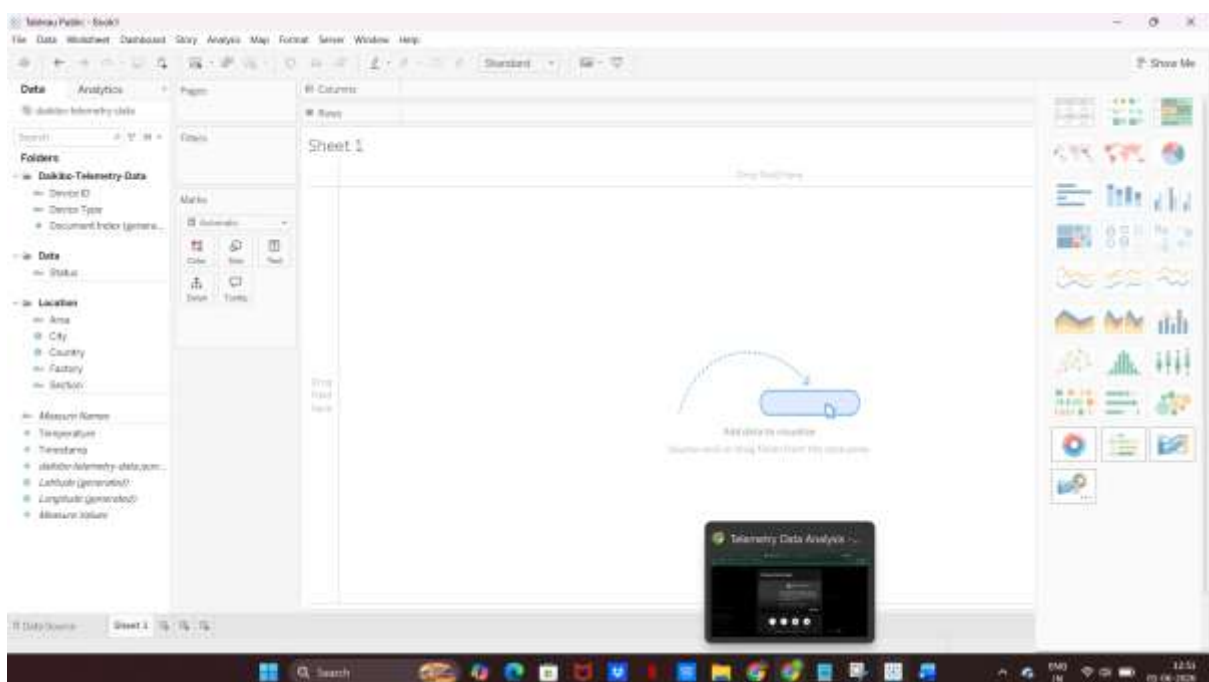
"Double check that you've checked all Schema levels when you import your data."

So you should have 3 checkmarks:

- daikibo-telemetry-data.json
- data
- location

Then click:

✓ OK



Great! 🎉 The data has loaded correctly. I can see:

- 160,704 rows ✓
- Fields like Status, Temperature, Device ID, Device Type ✓

Now let's build the assignment.

Step 1: Open Worksheet

Click Sheet 1 at the bottom left (orange tab).(sheet 1 name → Down Time per Factory)

Step 2: Create the "Unhealthy" Calculated Field

1. In the left Data pane, right-click in an empty area.
2. Select Create → Calculated Field.
3. Name it:

Unhealthy

4. Enter this formula:

```
IF [Status] = "unhealthy" THEN 10  
ELSE 0  
END
```

5. Click OK.

Then build the first chart

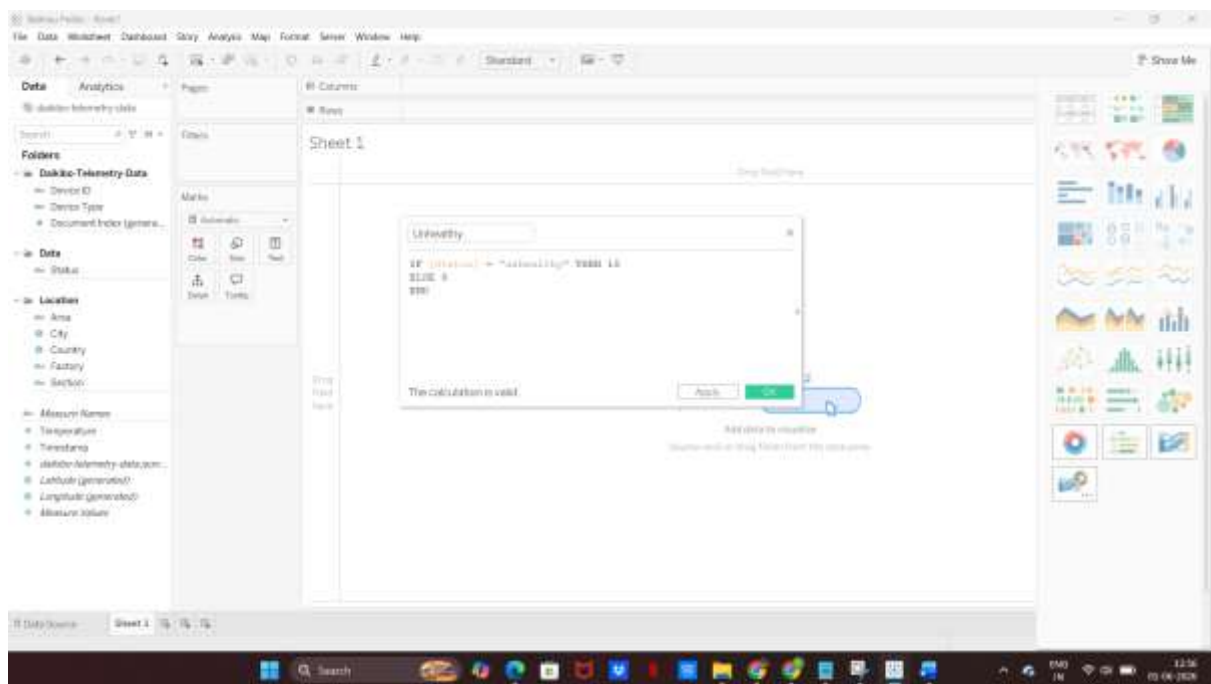
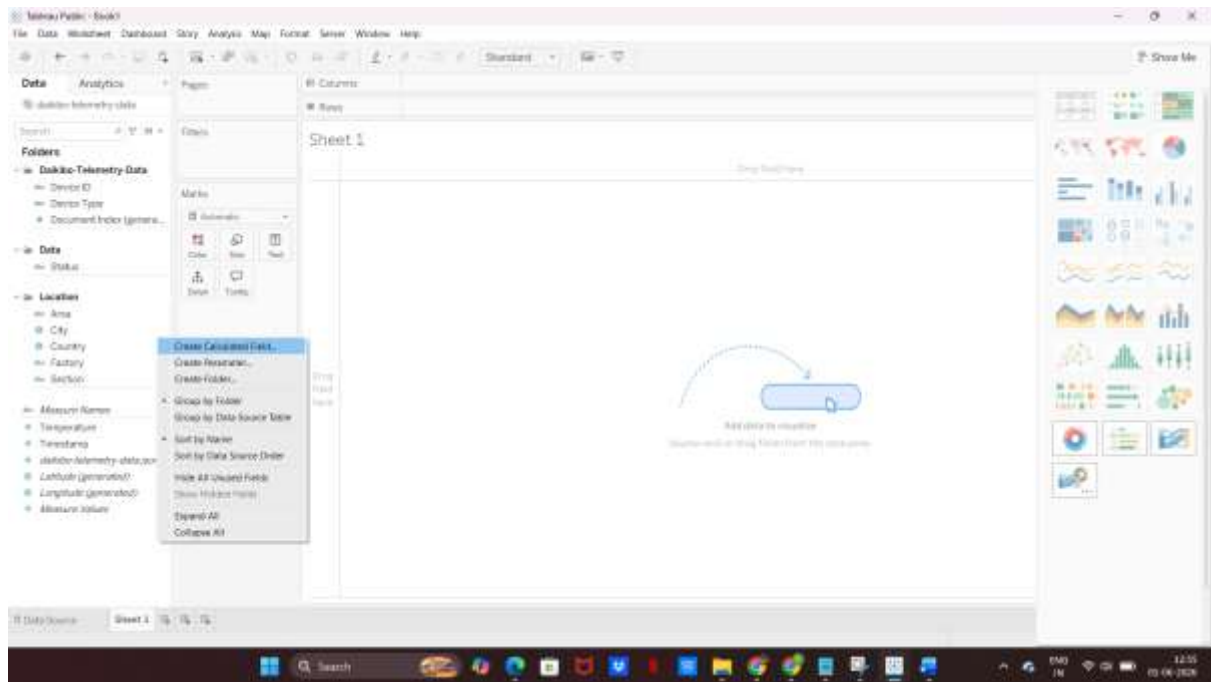
From the left panel:

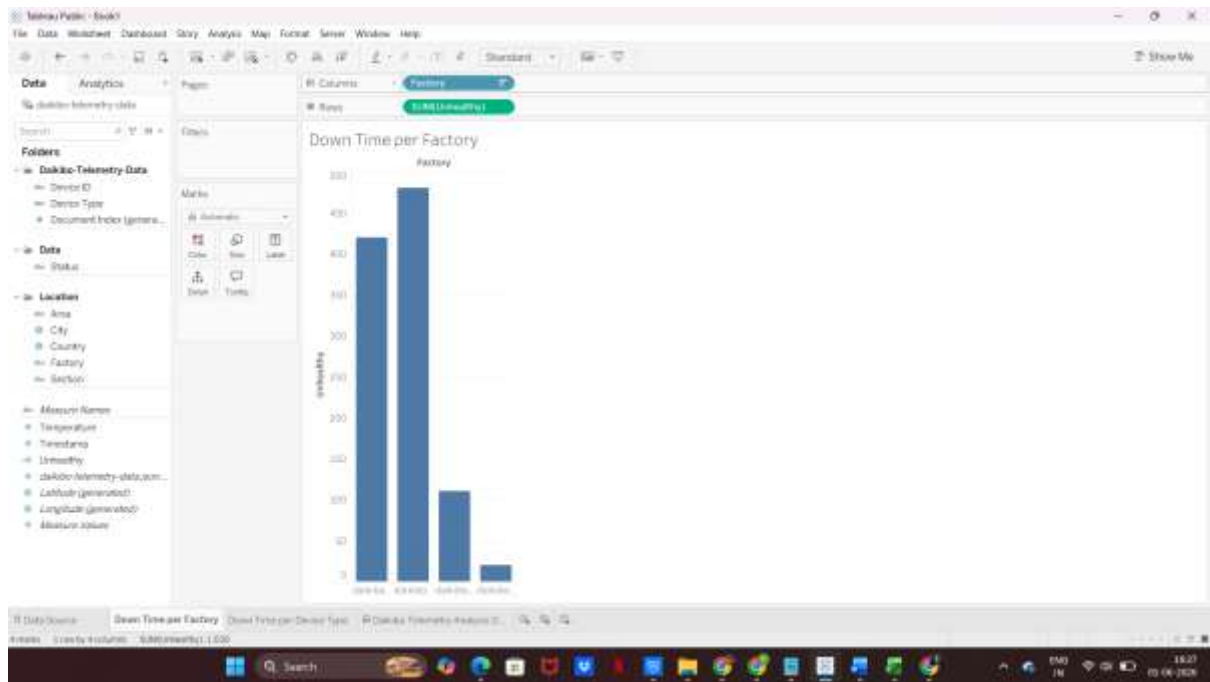
1. Drag Factory → columns
2. Drag Unhealthy → rows

You should immediately see a horizontal bar chart.

If Tableau shows AGG(Unhealthy) or SUM(Unhealthy), that's fine.

3. Click the Sort Descending icon on the toolbar.





Step 6: Create Second Worksheet

Click the **New Worksheet** icon (small sheet with +) next to the tabs at the bottom.

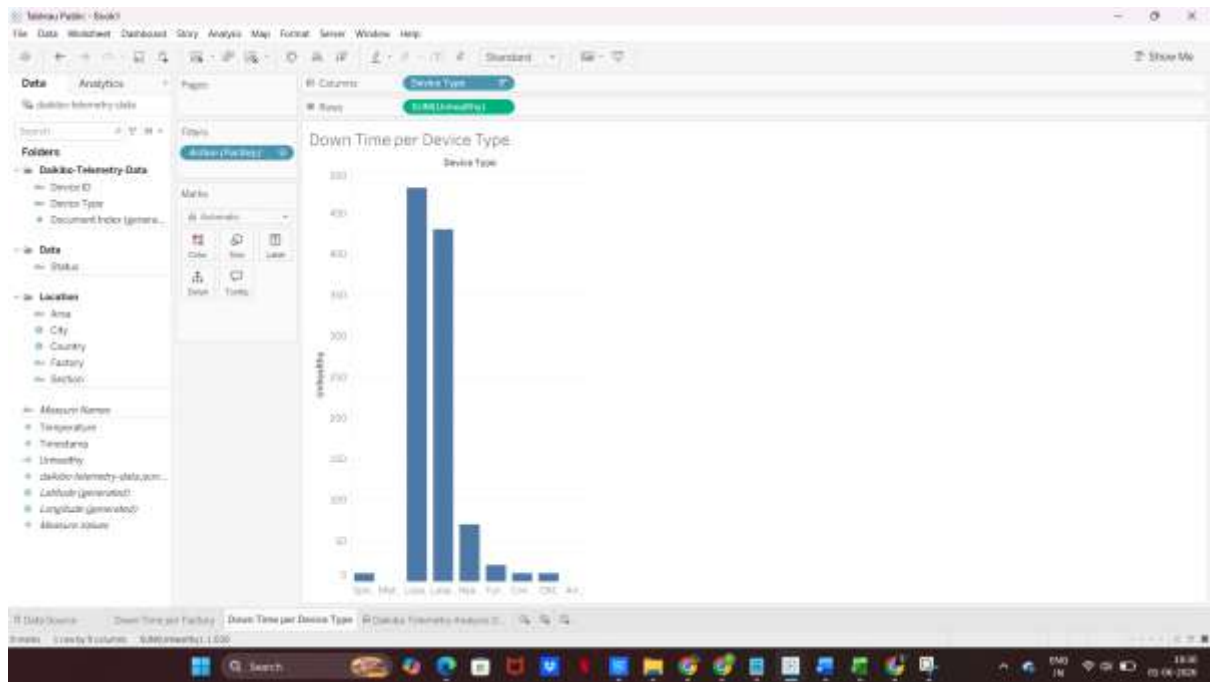
Step 7: Build "Down Time per Device Type"

1. Drag **Device Type** → **columns**
2. Drag **Unhealthy** → **rows**

You will get a bar chart showing downtime by machine type.

3. Rename the sheet:

Down Time per Device Type



Step 8: Create Dashboard

At the bottom, click the **New Dashboard** icon (looks like a grid).

OR

Menu:

Dashboard → New Dashboard

Step 9: Add Both Sheets

On the left side, drag:

First

Down Time per Factory

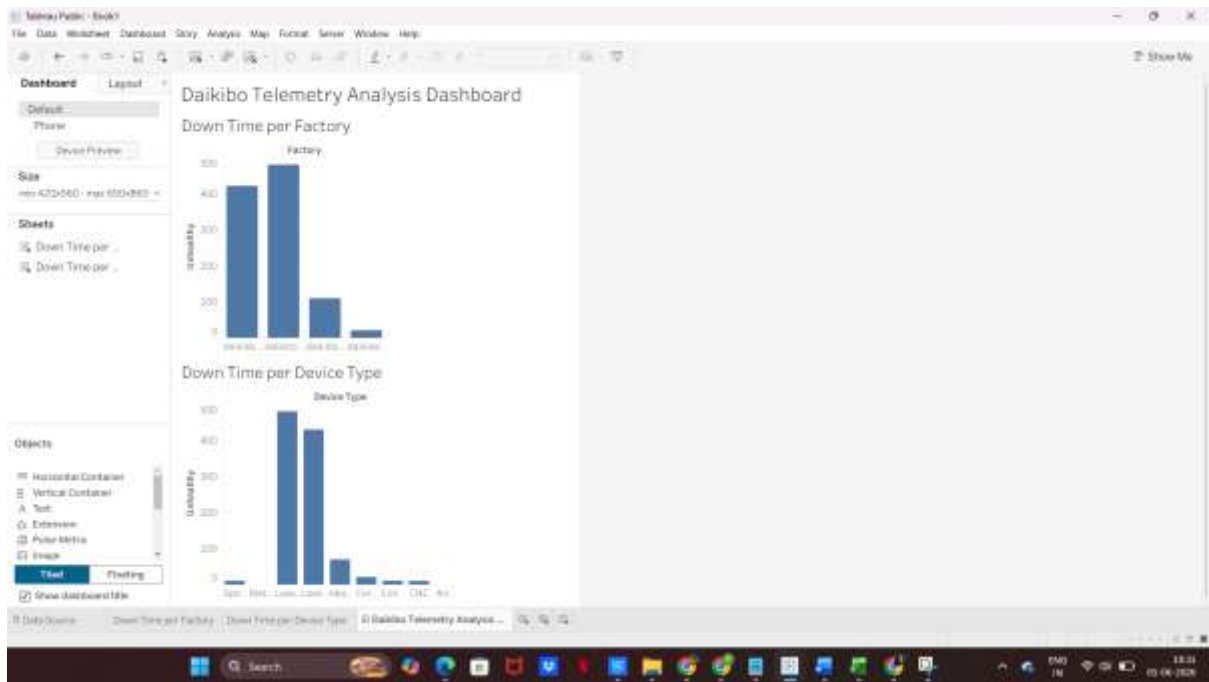
onto the dashboard.

Then

Down Time per Device Type

onto the dashboard.

Place them side-by-side.



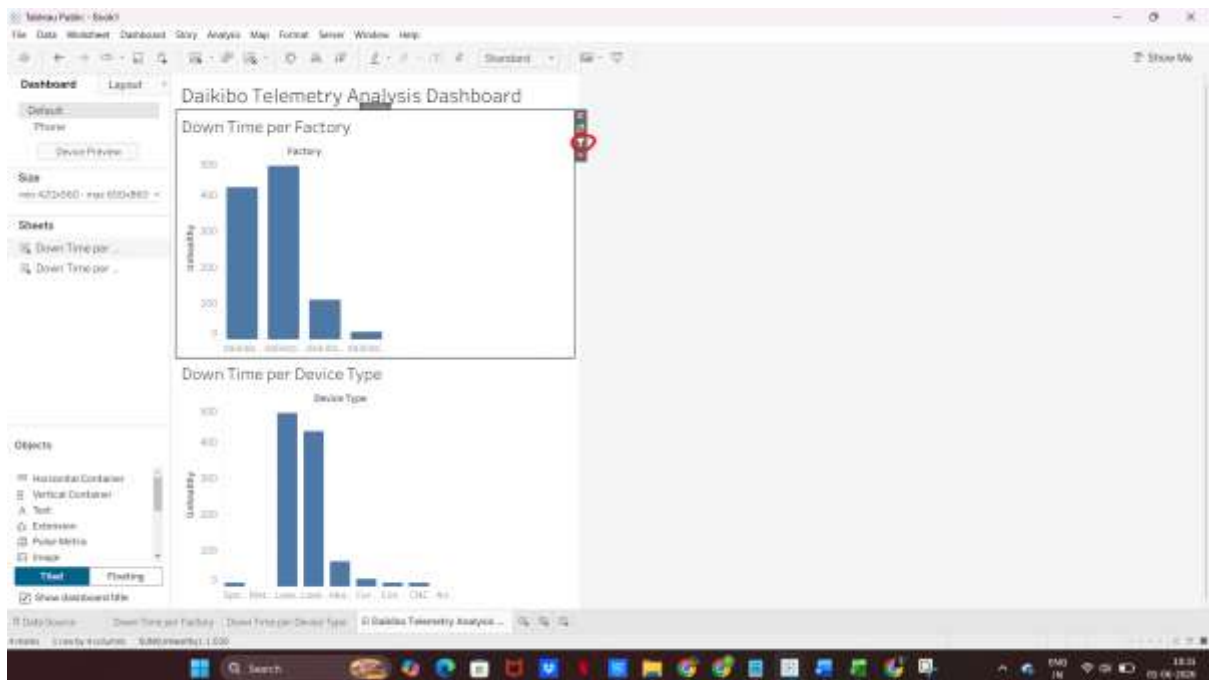
Step 10: Make Factory Chart a Filter

Click the small ▼ on the top-right corner of the **Down Time per Factory** chart.

Choose:

Use as Filter

A funnel icon should appear.



Final Dashboard Analysis – Daikibo Telemetry Data

Objective

Analyze one month of machine telemetry data (May 2021) from four Daikibo factories to identify:

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Dashboard Findings

1. Factory with the Highest Downtime

Factory	Total Downtime (Minutes)
Daikibo Factory Seiko	480
Daikibo Shenzhen	420
Daikibo Factory Meiyo	110
Daikibo Berlin	20

Finding:
Daikibo Factory Seiko (Osaka, Japan) recorded the highest machine downtime, totaling 480 minutes.

2. Machine Type with Highest Downtime in Seiko

After filtering the dashboard to Daikibo Factory Seiko, the machine breakdown showed:

Device Type	Downtime (Minutes)
LaserWelder	480
Others	0 or negligible

Finding:
The LaserWelder machine type accounted for the highest downtime in the Seiko factory.

Business Interpretation

- The Seiko factory experienced the greatest operational disruption among all factories.

- The LaserWelder was responsible for most of the downtime observed in Seiko.
 - Frequent LaserWelder failures may have impacted production efficiency and output.
 - Maintenance teams should prioritize investigating the LaserWelder equipment in the Seiko factory to reduce future downtime.
-

Conclusion

Based on the Tableau dashboard analysis:

- ✓ **Factory with most downtime: Daikibo Factory Seiko (Osaka, Japan)**
- ✓ **Machine with most downtime: LaserWelder**
- ✓ **Total downtime: 480 minutes (8 hours)**

This analysis helps Daikibo identify where maintenance efforts should be focused to improve factory performance and reduce machine failures.